

ANTÓN GÓMEZ LÓPEZ

Artificial Intelligence Engineer - Computer Engineer - Mathematician

 [antongomez.github.io/portfolio](https://github.com/antongomez)

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SUMMARY

AI Engineer & Researcher specialising in Deep Learning and Computer Vision, with a dual foundation in Computer Engineering and Mathematics.

Experienced in building and adapting deep learning architectures in PyTorch for complex computer vision tasks under data-scarce and class-imbalanced conditions.

Adapted a deep learning architecture achieving the highest average AA (90.4%) across 8 benchmark datasets, outperforming ViT and four other baselines; manuscript in preparation for a peer-reviewed journal. End-of-Degree Award recipient (top 4 out of 300+ graduates in Galicia).

EDUCATION

- | | | |
|---------------|--|--------------------------------------|
| 09/24 - 02/26 | Master's Degree in Artificial Intelligence
• Relevant Courses: Natural Language Understanding (TensorFlow), Machine Learning, Computer Vision (TensorFlow), Intelligent Robotics, AI Project Management, Deep Learning (Tensorflow), AI in Big Data Environments (PySpark)
• Grade: 9.28/10
• Courses finished with honours: 3
• Final Thesis: <i>Adaptation of StyleGAN for Processing Multidimensional Remote Sensing Images</i> | University of Santiago de Compostela |
| 09/18 - 07/24 | Bachelor's Degree in Computer Engineering
• Relevant Courses: Algorithms and Data Structures (programming language C), Computer Architecture, Object Oriented Programming (Java), Formal Languages and Automata Theory, Intelligent Systems (Python), Cybersecurity
• Grade: 9.24/10
• Courses finished with honours: 9
• Final Thesis: <i>Effective Adaptation of GANs for Multidimensional Remote Sensing Images</i> | University of Santiago de Compostela |
| 09/18 - 07/24 | Bachelor's Degree in Mathematics
• Relevant Courses: Linear and Multilinear Algebra, Numerical Methods in Optimization and Differential Equations, General Topology, Probability and Statistics, Regression Models and Multivariate Analysis (programming language R)
• Grade: 8.94/10
• Courses finished with honours: 8
• Final Thesis: <i>Classification Methods and Classifier Ensembles in Supervised Learning.</i> | University of Santiago de Compostela |

EXPERIENCE

- | | | |
|-------------------------------------|--|--|
| 02/25 - 02/26
scholarship holder | AI Research Scholar, Computer Vision
• Adapted a deep learning architecture for imbalanced, data-scarce classification by extending the discriminator with a fully connected classification head and training end-to-end in PyTorch on multidimensional data from scratch.
• Benchmarked against 5 architectures (CNN, ResNet, ViT, and two GAN-based baselines) across 8 independent datasets; achieved the highest average AA (90.4%) and the lowest cross-dataset variance (± 4.2 pp vs ± 8.3 pp for ViT).
• Publication: Manuscript in preparation for a top-tier peer-reviewed journal. | Research Center in Intelligent Technologies (CiTIUS) |
| 09/23 - 12/23 | Research Assistant, Machine Learning
• Implemented multiclass classification support in the open-source <i>caretEnsemble</i> R package, extending the existing codebase to integrate the team's novel statistical ensemble research.
• Wrote unit tests covering the new classification pipeline and shipped the contribution through the GitHub pull request workflow with maintainer code review. | Galician Centre for Mathematical Research and Technology |
| 06/23 - 08/23 | Intern, Machine Learning
• Optimized HuggingFace Transformers inference in Python, achieving a 2.5× throughput gain through Distributed Data Parallel (DDP) across nodes.
• Benchmarked Ray and Dask for distributed workload scheduling, evaluating scalability trade-offs to inform the team's infrastructure decisions. | Gradiant, Galician R&D Center in Advanced Telecommunications |

07/22 – 08/22
scholarship holder

Intern, Natural Language Processing

Research Center in Intelligent Technologies (CiTIUS)

- Trained and evaluated a Galician–Spanish neural machine translation model using OpenNMT, achieving a BLEU score of 64.3% under low-resource conditions.
- Built data processing and evaluation pipelines for reproducible benchmarking across resource-constrained language pairs.

AWARDS

05/25

End-of-Degree Award

Government of Galicia

Selected as one of 4 recipients among over 300 Computer Engineering students in Galicia, Spain, for having one of the top academic records in the 2024 graduating cohort.

2024

Additional awards

Best Free Software University Project, 2nd prize (CiXUG); Linguistic Quality Award for Final Degree Projects (USC).

PROJECTS & COMPETITIONS

2022 – 2025

Hackathons

HackUDC, HackUPC

- Won awards at 3 of 4 hackathons entered: 2nd place at HackUDC 2025 (300+ participants) and 2 prizes at HackUDC 2024 (150+ participants), including Best Open Source Project and the BizAway industry challenge.
- Built AI-powered outfit recommendations (Wearvana, FastAPI + React), graph-based carbon-optimal travel routing (EcoViaggio, networkx + FastAPI), and NP-complete expense minimisation (OpenSplitBot, Rust + Python) under 30-hour constraints.

2023, 2024

Ada Byron Competitive Programming

ada-byron.es

- Qualified for the national stage in both editions entered; ranked 13th/34 (2023) and 11th/42 (2024) nationally among university teams.

LANGUAGES

English - C1 Advanced (Cambridge Certificate), **Spanish** - native, **Galician** - native

07/22 - 08/22

English course

Limerick Language Center

A three-week stay in Limerick taking a B2-level English.

VOLUNTEERING

02/26

Hackathon Organizer

A Coruña, Galicia, Spain

Served as an organizer of HackUDC 2026, the second largest university hackathon in Spain (over 400 participants, €8k in prizes), coordinating sponsors, defining challenges and prizes, and ensuring educational value for participants.

07/21

Environmental volunteering

Baixa Limia Natural Park, Galicia, Spain

Performed cleaning tasks, adapted areas, and monitored native forest regeneration zones.

06/21 - 09/21

Cultural volunteering

Cambre, Galicia, Spain

Promoted the English Way through a volunteer initiative supported by the Cambre City Council, providing information about local heritage and assisting pilgrims.

06/19 - 09/19

Tourist Entertainer

Praga, Czech Republic

Collaborated with a hostel in Prague (Hostel Advantage) as a volunteer, performing daily tasks at the hostel and organizing entertainment and social activities within the hostel.